

End of semester exam

A rural bank with 5 branches scattered across Ghana acquired the IP address 195.143.208.133/25 from an ISP.

- a. What class is the IP address?
- b. What subnet is the address on?
- c. What is the NetID of the address?
- d. What is the HostID of the address?
- e. What is the SubnetID of the address?
- f. What is the site address?
- g. Give the address in its BITWISE notation.
- h. What does the /25 means?
- i. Give the dotted decimal notation of the /25 and also give bitwise notation equivalent.
- j. Design a suitable network for the organisation ensuring the most efficient usage of addresses.
- k. List all the SubnetID's
- l. Give the subnet address that this IP address resides on 195.143.208.133/28
- m. What is the site address for the given address in 'l' above
- n. What is the NetID for the given address in 'l' above
- o. What is the HostID for the given address in 'l' above
- p. What is the SubnetID for the given address in 'l' above
- q. Draw the network diagram that can be used for the implementation of the organisation network
- r. Explain the functionality of each of the devices in your network diagram and justify their purpose in your diagram
- s. How will you ensure Internet connectivity for host at each of the 5 branches.
- t. Explain type(s) of connections in your network diagram
- u. Explain the protocols for devices communication in your network diagram
- v. Explain protocol for 3 applications communication on your network
- w. Explain the addressing mechanism in the movement of packets from one branch to the other
- x. Assuming each branch has 8 workstations and a network printer, how many collision domains has your network diagram
- y. As follow-up question to 'x', how many broadcast domains has your network diagram?

z. Write a convincing summary statement to convince management why they should invest money to deploy your network diagram. Your statement must clearly spell out the advantages of the network that the organisation stands to gain if they implement it.

Submission deadline is 6pm on Monday 22nd June, 2020

Submission to my email

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